

Configuration Management Plan

Mangrove Monitoring System — Deep Learning-Based Detection of Mangrove Gain and Loss in Brgy. Dulao, Aringay, La Union

Item ID	Item / File Name	Category	Filepath / Repository Path	Primary Owner	Version	Lifecycle Status
CI-01	app.py	Source Code	/SourceCode/app.py	Nash Francis Caluza	v1.0	● Active
CI-02	templates/ (base, index, annual maps, change detection, statistics, model_evaluation, report)	Source Code	/SourceCode/templates/	Nash Francis Caluza	v1.0	● Active
CI-03	style.css	Source Code	/SourceCode/static/css/style.css	Nash Francis Caluza	v1.0	● Active
CI-04	common.js, map_annual.js, map_change.js, charts.js	Source Code	/SourceCode/static/js/	Nash Francis Caluza	v1.0	● Active
CI-05	config.py	Source Code	/SourceCode/preprocessing/config.py	Nash Francis Caluza	v1.0	● Under Development
CI-06	gee_export.py	Source Code	/SourceCode/preprocessing/gee_export.py	Nash Francis Caluza	v1.0	● Under Development
CI-07	make_patches.py	Source Code	/SourceCode/preprocessing/make_patches.py	Nash Francis Caluza	v1.0	● Under Development
CI-08	rasterize_labels.py	Source Code	/SourceCode/preprocessing/rasterize_labels.py	Nash Francis Caluza	v1.0	● Under Development
CI-09	unet_model.py	Source Code	/SourceCode/preprocessing/unet_model.py	Nash Francis Caluza	v1.0	● Under Development
CI-10	train.py	Source Code	/SourceCode/preprocessing/train.py	Nash Francis Caluza	v1.0	● Under Development
CI-11	evaluate.py	Source Code	/SourceCode/preprocessing/evaluate.py	Nash Francis Caluza	v1.0	● Under Development
CI-12	change_detection.py	Source Code	/SourceCode/preprocessing/change_detection.py	Nash Francis Caluza	v1.0	● Reserved
CI-13	preprocessing README.md	Source Code	/SourceCode/preprocessing/README.md	Nash Francis Caluza	v1.0	● Under Development
CI-14	README.md (root)	Documentation	/README.md	Lyka Vejano	v1.0	● Revised (Post-Proposal)
CI-15	01_Abstract.md	Documentation	/Documentation/01_Abstract.md	Lyka Vejano	v1.0	● Reserved
CI-16	02_Methodology_Summary.md	Documentation	/Documentation/02_Methodology_Summary.md	Lyka Vejano	v1.0	● Reserved
CI-17	03_Data_Acquisition_Protocol.md	Documentation	/Documentation/03_Data_Acquisition_Protocol.md	Lyka Vejano	v1.0	● Reserved
CI-18	04_Labeling_Guidelines.md	Documentation	/Documentation/04_Labeling_Guidelines.md	Lyka Vejano	v1.0	● Reserved
CI-19	05_User_Manual.md	Documentation	/Documentation/05_User_Manual.md	Lyka Vejano	v1.0	● Reserved

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CI-20	06_Defense_Presentation.pdf	Documentation	/Documentation/06_Defense_Presentation.pdf	Lyka Vejano	v1.0	● Reserved
CI-21	07_Thesis_Manuscript.pdf	Documentation	/Documentation/07_Thesis_Manuscript.pdf	Lyka Vejano	v1.0	● Reserved
CI-22	.gitignore	Dependencies & Env	/.gitignore	Reymark Boado	v1.0	● Active
CI-23	requirements.txt	Dependencies & Env	/Dependencies and Environment/requirements.txt	Reymark Boado	v1.0	● Active
CI-24	environment.yml	Dependencies & Env	/Dependencies and Environment/environment.yml	Reymark Boado	v1.0	● Active
CI-25	python-version.txt	Dependencies & Env	/Dependencies and Environment/python-version.txt	Reymark Boado	v1.0	● Active
CI-26	SETUP_GUIDE.md	Dependencies & Env	/Dependencies and Environment/SETUP_GUIDE.md	Reymark Boado	v1.0	● Active
CI-27	RUNBOOK.md	Dependencies & Env	/Dependencies and Environment/RUNBOOK.md	Reymark Boado	v1.0	● Active
CI-28	training_pipeline.ipynb	Dependencies & Env	/Dependencies and Environment/colab_notebooks/training_pipeline.ipynb	Reymark Boado	v1.0	● Under Development
CI-29	colab notebooks README.md	Dependencies & Env	/Dependencies and Environment/colab_notebooks/README.md	Reymark Boado	v1.0	● Active
CI-30	DATA_SCHEMA.md	Data & Schema	/Data and Schema/DATA_SCHEMA.md	Nash Francis Caluza	v1.0	● Active
CI-31	data_manifest.md	Data & Schema	/Data and Schema/data_manifest.md	Nash Francis Caluza	v1.0	● Active
CI-32	reference_sources.md	Data & Schema	/Data and Schema/reference_sources.md	Nash Francis Caluza	v1.0	● Active
CI-33	study_area_boundary.geojson	Data & Schema	/Data and Schema/samples/study_area_boundary.geojson	Nash Francis Caluza	v1.0	● Active
CI-34	sample_labels.geojson	Data & Schema	/Data and Schema/samples/sample_labels.geojson	Nash Francis Caluza	v1.0	● Active
CI-35	sample_patch_index.csv	Data & Schema	/Data and Schema/samples/sample_patch_index.csv	Nash Francis Caluza	v1.0	● Active
CI-36	sample_metrics.example.json	Data & Schema	/Data and Schema/samples/sample_metrics.example.json	Nash Francis Caluza	v1.0	● Active
CI-37	MODEL_CARD.md	Model Artifact	/Model/MODEL_CARD.md	Nash Francis Caluza	v1.0	● Under Development
CI-38	metrics.example.json	Model Artifact	/Model/metrics.example.json	Nash Francis Caluza	v1.0	● Active
CI-39	metrics.json	Model Artifact	/Model/metrics.json	Nash Francis Caluza	v1.0	● Reserved
CI-40	weights/	Model Artifact	/Model/weights/	Nash Francis Caluza	v1.0	● Under Development

Status Legend

-  **Under Development** — active coding or training in progress; item not yet through a full revision cycle.
-  **Revised (Post-Proposal)** — item has been updated to incorporate the evaluators' comments from the thesis proposal defense; revision round complete, awaiting re-baselining / adviser re-approval.
-  **Active** — baseline approved and operational for development/production.
-  **Draft** — initial documentation undergoing review.
-  **Reserved** — allocated placeholder directory/file for future artifacts.

Change Control Protocol

- Any modification to a CI increments its version and updates its Lifecycle Status in this register.
- The commit message must reference the CI ID (e.g., "CI-11: revise evaluation split persistence — v1.1").
- Status transitions — Development path:  Reserved →  Under Development →  Revised (Post-Proposal) →  Active
- Status transitions — Documentation path:  Reserved →  Draft →  Revised (Post-Proposal) →  Active
- An item returns to  Under Development whenever rework opens again.
- Items updated in response to thesis proposal defense panel comments must be marked  Revised (Post-Proposal) once the revision round for that item is complete, and the revision round must be logged in Version History.
- Large binary artifacts (rasters, patches, weights) are tracked via the Drive locations recorded in Data and Schema/data_manifest.md; only their register entries change in Git.